11 Push and Pull

We push or pull an object to move it, stop it, change its shape, or change its direction of movement. A push or a pull acting on an object is called force. Force is required in our various day-to-day activities.

Learn about

- Push and pull
- Force
- Types of forces

Push and Pull

We push or pull different objects daily.

While **pushing** an object we hold the object and use our strength to move it forward. The object moves away from us.

Pushing a mower

Some examples of push are:

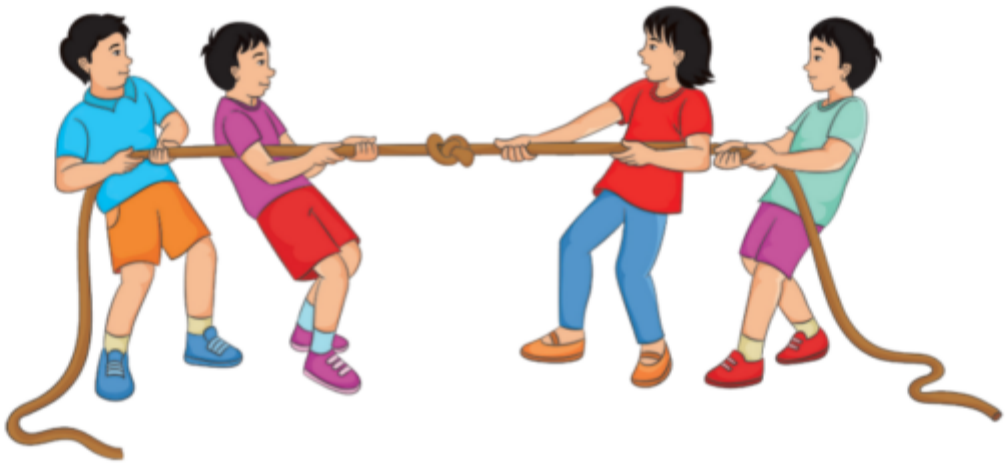
- Closing an open drawer
- Pushing striker while playing carrom
- Placing books on the bookshelf
- Kicking the football
- Pushing a mower

Pulling is the opposite of pushing. While pulling an object, we hold it and use our strength to move it towards us. The object moves closer to us.

Some examples of pull are:

- Opening a drawer
- Pulling up socks
- Taking a book from the bookshelf
- Pulling rope while playing tug of war
- Pulling a rubber band
- A bullock pulling a cart

Pulling a drawer



Pulling a rope while playing tug of war

▶ Force

A push or a pull is called ***force****.* Force help us to do various activities. Some of them are listed in the table given below.

Move an object	Stop a moving object	Change the direction of movement of an object	Change the shape and size of an object
An object can be moved by applying force on it.	A moving object can be stopped by applying force on it.	The direction of a moving object can be changed by applying force on it.	The shape of an object can be changed by applying force on it.
Example: A man moves a broken car by pushing it.	*Example:* A kid stops a moving ball by catching it.	*Example:* A batsman changes the direction of a ball by hitting it with his bat.	*Example:* A lump of clay can be moulded into different shapes.

▶ Types of Forces

There are different types of forces. Some of them are discussed here.

Gravitational Force

A ball when thrown upwards, comes down after covering some distance. This happens due to a force called gravity. *The force that pulls every object downwards, towards the centre of the Earth, is called* ***force of gravity*** *or* ***gravity****.*

The first person who discovered the force of gravity was Sir Isaac Newton.

The force of gravity pulls the ball down.

Fact File

Planets keep moving in their orbits, and do not change their path due to the gravitational pull of the Sun.

The direction of frictional force is always opposite to the direction of the moving object.

Frictional Force

The force that acts whenever any two objects rub together or come in contact is called ***frictional force*** *or* ***friction****.* Frictional force always acts in the direction opposite to the direction of the moving object. Friction slows down a moving object or completely stops it from moving.

Questions

Mention the type of force (pull or push) applied in each case.

1. To close a drawer .....
2. To move a table forward .....
3. To draw a chair towards us .....
4. To kick a football .....
5. To pick a book from the bookshelf .....

Magnetic Force

A magnet has the capacity to attract certain metals towards itself. *The force with which a magnet attracts certain metals, such as iron and nickel, towards itself is called* ***magnetic force***. A magnet can also attract or repel another magnet with its magnetic force. A magnet has two poles: North Pole (marked as N) and South Pole (marked as S).

Iron nails are attracted to a magnet due to magnetic force.

Muscular force is applied while lifting a heavy weight.

Muscular Force

In the human body, movement is possible with the help of muscles. Running, walking, standing, sitting, pushing, pulling, and carrying things are different kinds of work that we can do with the help of our muscles. *The force exerted by the muscles to do work is called* ***muscular force***.

Think and Discuss

An athlete or a young child: who will possess more muscular force?

Wrap Up

- While pushing an object we hold the object and use our strength to move it forward. The object moves away from us.
- While pulling an object we hold it and use our strength to move it towards us. The object moves closer to us.
- Force is a push or pull acting on an object.
- Force can move an object, stop a moving object, change the direction of movement of an object, and change the shape of an object.
- Force can be of many types, such a gravitational force, frictional force, magnetic force, and muscular force.

Exercises

SECTION I

REMEMBERING UNDERSTANDING APPLYING

A Choose the correct option.

1. Which of the following is an example of pull?
   a. Pressing a remote control button b. Opening a drawer
   c. Nailing d. Writing with a pencil
2. Which of the following actions does not require force?
   a. Watching a movie b. Writing
   c. Throwing a ball d. Opening the door
3. Which of the following is NOT an effect of force on an object?
   a. Change of direction of movement of an object
   b. Making a stationary object move
   c. Changing the colour of an object
   d. Changing the shape of an object
4. Which of the following is the reason for a coconut falling from a coconut tree?
   a. Solar energy b. Friction c. Gravity d. Magnetic energy
5. The force exerted by our muscles is ..... force.
   a. muscular b. frictional c. magnetic d. gravitational
6. The force with which a magnet attracts other substances is ..... force.
   a. magnetic b. electrical c. gravitational d. muscular

B Assertion and Reasoning questions.

1. **Assertion (A):** Taking your books out of your bag is an example of push.
   Reason (R): A force can move an object.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True
2. **Assertion (A):** When you try to push a table, friction acts in opposite direction.
   Reason (R): The direction of frictional force is in the direction of movement of object.
   a. Both A and R are True b. Both A and R are False
   c. A is True and R is False d. A is False and R is True

C Name the following.

1. The force involved when we use our strength to move an object .....

2. The force due to which a magnet pulls a nail towards itself .....
3. A push or pull acting on an object .....
4. The force due to which a mango falls from the tree .....
5. The force that helps you to write .....
6. Moving an object away from us .....

SECTION II

APPLYING ANALYSING

D Short answer questions.

1. Define force.
2. Name the four types of forces.
3. Name two metals that are attracted by a magnet.
4. Differentiate between push and pull.
5. Name the force involved while running.

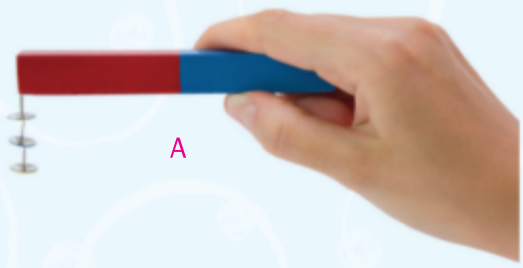
E Long answer questions.

1. Explain why a ball rolling on a ground, comes to rest after some time.
2. Explain the effects of force on an object with suitable examples.
3. Ravi's parents bought a new study table for him. He is trying hard to change its position but is unable to do so. Identify the force acting here and define it.
4. Explain the force that helps you carry your school bag.

Picture Study

TP ANALYSING

Look at the picture and answer the following questions.



1. Name the object **A** shown in the picture.
   .....
2. Name a metal attracted to object **A**.
   .....
3. Which force is responsible for the attraction of the metal to object **A**?
   .....
4. Write two features of object **A**.
   a. .....
   b. .....

My Learning Corner

A Think about

1. Suppose a drawer of your study table is closed and you want to open it. Which force will you apply—pull or push?
2. If the floor is smooth, will the force of friction be more or less? Give reasons.

B Try out

1. List down the important rules of cricket, football, or any other outdoor game of your choice. Name the type of force applied in that game.
2. Suppose your parents have given you the task of cleaning up your study room. For this, you have to arrange things and keep them at their proper places—all stationery items in the drawer of the study table, books on the bookshelf, and arrange the study table properly. You may also have to do a little dusting and cleaning of the floor. Make a list of all the 'pull' and 'push' activities that you would be doing during this cleaning task. Write them down in your Science Scrapbook in a tabular form.
3. Perform this group activity in your classroom. The whole class can be divided into groups of four. Each group will demonstrate an activity based on any one force: Gravitation/Friction/Magnetic/Muscular. You can consult elders, teachers, or the Internet for it. Collect things required for the activity beforehand and perform your best as a team.

Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• define push and pull.	
• define force.	
• explain and demonstrate the different types of forces through various activities.	

Worksheet

Observe each picture carefully, and mention the type of force applied in each case.

	Pictures	Type of force applied
1.		....
2.		....
3.		....
4.		....